The normal model for options on swap rates has become the industry standard, eclipsing Black's model. That is, it is more popular nowadays to employ a normal model for the pricing of options on swap rates than it is to assume lognormality and employ Black's model. Traders in the market are more likely to speak of vol in normal terms than in lognormal terms. That being said, lognormal implied vols remain a benchmark, at least for at-the-money options.

5.5 Other Models Used to Price Interest Rate Options

We have barely begun to scratch the surface in terms of the various theoretical models that are widely used to price interest rate options. The lognormal and normal models are both powerful paradigms, but neither accurately describes the dynamics of skew and options prices completely. Depending on supply and demand effects in the market, there are times when the skew pattern observed from interest rate option prices suggests a lognormal model is more appropriate, times when a normal model is more appropriate, times when interest rates seem to be behaving as a blend of a normal and lognormal process, and times when rates are behaving altogether differently.

There is a vast literature on alternative models to price interest rate options, and the industry has adopted a number of other models as favorites over the course of time. A full description of these models is beyond the scope of this text, but here are a couple of models that have been popular:[39]

1. **CEV**. The *Constant Elasticity of Variance* (*CEV*) model is one in which volatility is assumed to be of the form

   $$\sigma F^{\beta},$$

   where F is the underlying forward swap rate, β is a constant known as the *CEV coefficient*, and σ is a constant. The CEV model belongs to a class of models known as *local volatility models*, meaning it is assumed that vol depends on the level of rates. This is a more general model than either the normal model or the lognormal model. The CEV model collapses to the normal model when $\beta = 0$ and to the lognormal model

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> sleeps under the blanket of freedom I provide, then questions the manner in which I provide it. I prefer you said thank you, and went on your way.

Excerpts from the movie *A Good Few Men*, released on December 11, 1992. Used by permission of Sony Pictures Entertainment.

[39]See Brigo and Mercurio (2006) for more details of alternative interest rate option models.

when $\beta = 1$. A value of $\beta < 0$ is consistent with a *supernormal skew*, as depicted in Figure 5.9.

2. **SABR**. The *SABR model* (standing for "Stochastic Alpha, Beta, Rho"—the three parameters in the model that need to be estimated) is a generalization of the CEV model in which the volatility parameter σ itself is assumed to be stochastic. This model collapses to the CEV model when one of the parameters (i.e., alpha) is set to zero.

Despite the doubt cast over the appropriateness of the lognormal model, and the proliferation of alternative models including the normal model, we will continue to employ Black's model unless otherwise stated. Moreover, we will assume flat skew (i.e., varying the strike of an option does not change its implied vol, all else equal) unless explicitly stated otherwise.

5.6 Bermudan Swaptions

A *Bermudan* or *multi-European swaption* is a swaption that can be exercised on multiple dates.[40] Consider, for example, a 6-month into 5-year Bermudan receiver swaption with semiannual exercise. This swaption gives the holder the right to receive fixed in a 5-year swap six months from now. If the holder does not exercise in six months, the holder has the right to exercise every six months thereafter into the remaining term of the underlying swap. If, for example, the holder does not exercise at the first exercise date in six months, nor at the next exercise date six months thereafter, but does exercise at the third exercise date (which is 1.5 years from today), then the holder will receive fixed at the strike rate for the remaining term of the underlying swap, which in this case is four years. Once an exercise has been made, there is no further opportunity to exercise again. Table 5.7 shows the possible exercise dates and underlying swaps associated with this Bermudan swaption. The holder of the 6-month into 5-year Bermudan swaption with semiannual exercise may elect to exercise on at most one of the exercise dates specified in Table 5.7 and upon exercise will enter into the corresponding remaining term of the swap.

One thing to note right off the bat is that the 6-month into 5-year Bermudan swaption must be worth as much as any of the single European swaptions listed in Table 5.7.[41] And since the Bermudan must be worth at least

[40] This differs from an American swaption, which can be exercised continuously throughout some period of time. American swaptions are pretty unusual to see in the swaptions market, certainly as compared to Bermudans.

[41] If the Bermudan were worth less than a particular single European in this table, we could buy the Bermudan and sell the European, pocketing some cash today. We would hold the Bermudan until the exercise date of the European. If the European is exercised,

Exercise Date	Underlying Swap
6 months	5-year swap
1 year	4.5-year swap
1.5 years	4-year swap
2 years	3.5-year swap
2.5 years	3-year swap
3 years	2.5-year swap
3.5 years	2-year swap
4 years	1.5-year swap
4.5 years	1-year swap
5 years	6-month swap

Table 5.7: Possible Exercise Dates and Underlying Swaps Associated with a 6-Month into 5-Year Bermudan Swaption with Semiannual Exercise

as much as any of the particular European options in this table, it must be the case that the Bermudan is worth at least as much as whichever of these Europeans has the greatest value. This therefore places a meaningful lower bound on the price of Bermudan swaptions.

Example

Consider again our stylized example and suppose the swap curve today is as in Table 5.2. Suppose further that 1 into 2 European lognormal swaption vol is 20% and also that 2 into 1 European lognormal swaption vol is 20%. Consider now \$100mm of a 1 into 2 Bermudan payer swaption struck at 5%. This option gives the holder the right to pay fixed in a 2-year swap at a rate of 5% in one year and, if not exercised at this point, the right to pay fixed in a 1-year swap at a rate of 5% in two years. As shown in Figure 5.16, using Black's formula the value of \$100mm of a 1 into 2 European payer swaption struck at 5% is \$1,211,221, and the value of \$100mm of a 2 into 1 European payer swaption struck at 5% is \$876,266. Thus, we know that the 1 into 2 Bermudan payer swaption struck at 5% must be worth at least \$1,211,221. □

we would exercise the Bermudan as well, resulting in net zero cash flows. If the European is not exercised, we can possibly exercise the Bermudan at some later time, generating positive profits. Thus we have nonnegative cash flows in the future, which would be an arbitrage. So the Bermudan must be worth at least as much as any of the Europeans.

Maturity	Swap Rate (Ann 30/360)
1	5.01%
2	5.01%
3	5.01%
4	5.01%

Table 5.8: Swap Curve Prevailing in One Year in Example 1

5.6.1 Optimal Exercise of Bermudan Swaptions

At the expiry of a European swaption, optimal exercise depends solely on how the then-prevailing swap rate in the underlying swap compares with the strike. For example, the holder of a 1 into 5 European receiver swaption will exercise in a year if the 5-year swap rate in a year is less than the strike. For Bermudan swaptions, the optimal exercise decision on any exercise date may be complicated by the presence of additional options remaining in the Bermudan.

Example 1

Consider again our stylized example with today's swap curve as given in Table 5.2 and consider a \$100mm 1 into 2 Bermudan payer swaption struck at 5% that we are long. In this example, we will consider optimal expiry of this option in one year. Suppose that in one year, the swap curve is as in Table 5.8. Suppose further that one year from now 1 into 1 European swaption vol is 20%. Is it optimal to exercise our Bermudan payer swaption at this time?

Note that the 2-year swap rate prevailing at this time is 5.01% and is therefore above the 5% strike of our Bermudan payer swaption. If we exercise at this point, we will be paying fixed in a 2-year swap at a rate of 5% when the 2-year swap rate is 5.01%. Thus, the value of the swaption if we exercise today is equal to the present value of an annuity equal to $5.01\% - 5.00\% = 1$ bp of the notional, running for two years.[42] On an assumed notional of \$100mm, the value of early exercise is therefore the value of this annuity, which can be shown to be \$18,591.

[42] Upon exercise of the payer swaption, we will be paying 5% when the 2-year swap rate is 5.01%. As we discussed in Chapter 3, the value of this swap is the value of an annuity equal to the difference between the then-prevailing swap rate and the strike. Of course, upon exercise, we are not locking in the value of this annuity if the swap is physically settled. With physical settlement, we will actually enter into the underlying swap upon exercise and are exposed to changes in value of the swap as the market moves.

Swaption model		1	into	2		
Time	Swap rate	zc	Period	$DF(0,t)$		
1	5.000%	5.000%	0x1		d_1	0.5421
2	5.110%	5.113%	1x2	0.9051	d_2	0.3421
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4		$FN(d_1) - XN(d_2)$	0.006877
5	5.600%	5.635%	4x5		Payer swaption	1,211,221
					(in bps)	121.1
F	5.462%		A =	1.7613		
σ	20.0%					
T	1		Notional	100,000,000		
X	5.000%					

Swaption model		2	into	1		
Time	Swap rate	zc	Period	$DF(0,t)$		
1	5.000%	5.000%	0x1		d_1	0.6122
2	5.110%	5.113%	1x2		d_2	0.3294
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4		$FN(d_1) - XN(d_2)$	0.010235
5	5.600%	5.635%	4x5		Payer swaption	876,266
					(in bps)	87.6
F	5.712%		A =	0.8562		
σ	20.0%					
T	2		Notional	100,000,000		
X	5.000%					

Figure 5.16: Valuation of 1 into 2 European Payer and 2 into 1 European Payer

Swaption model		1	into	1		
Time	Swap rate	zc	Period	$DF(0,t)$		
1	5.010%	5.010%	0x1		d_1	0.1100
2	5.010%	5.010%	1x2	0.9069	d_2	-0.0900
3	5.010%	5.010%	2x3			
4	5.010%	5.010%	3x4		$FN(d_1)$ - $XN(d_2)$	0.004037
					Payer swaption	366,095
					(in bps)	36.6
F	5.010%		A =	0.9069		
σ	20.0%					
T	1		Notional	100,000,000		
X	5.000%					

Figure 5.17: Valuation of 1 into 1 European Payer Struck at 5% when the Swap Curve Is as in Table 5.8 and 1 into 1 Vol Is 20%

If, however, we do not exercise our Bermudan swaption, we will be long the residual option in the Bermudan, which is a 1 into 1 European payer swaption struck at 5%. As shown in Figure 5.17, the value of this swaption is $366,095. Since the value of the 1 into 1 payer far exceeds the value of the annuity obtainable upon immediate exercise, we are much better off forgoing exercise in a year. Note that even though the 2-year swap rate in one year, 5.01%, is greater than the strike, we optimally choose to not exercise. In this scenario, exercising the payer swaption that was slightly in-the-money did not justify giving up the additional option that we were long. □

Example 2

This example is identical to Example 1 except now we have a different swap curve prevailing in a year. Consider again our $100mm 1 into 2 Bermudan payer swaption struck at 5% that we are long. Suppose that in one year, the swap curve is as in Table 5.9. Suppose further that one year from now 1 into 1 European swaption vol is 20%, which was the level of 1 into 1 swaption vol assumed to be prevailing in one year in Example 1. Is it optimal to exercise our Bermudan payer swaption at this time?

Note that the 2-year swap rate prevailing at this time is 5.30% and is therefore significantly above the strike of our Bermudan payer swaption. If

Maturity	Swap Rate (Ann 30/360)
1	5.30%
2	5.30%
3	5.30%
4	5.30%

Table 5.9: Swap Curve Prevailing in One Year in Example 2

we exercise at this point, we will be paying fixed in a 2-year swap at a rate of 5% when the 2-year swap rate is 5.30%. Thus, the value of the trade if we exercise at this time is equal to the present value of an annuity equal to 5.30% − 5.00% = 30 bps of the notional, running for two years. On an assumed notional of $100mm, the present value of this annuity can be shown to be $555,461.

If however we do not exercise our Bermudan swaption, we will be long the residual option in the Bermudan, which is a 1 into 1 European payer swaption struck at 5%. As shown in the top half of Figure 5.18, the value of this swaption is $520,785. In this case, we are better off with immediate exercise since the value from immediate exercise, $555,461, is greater than the value of the remaining option if we do not exercise. □

Example 3

This example is identical to Example 2 except now we have a different level of prevailing 1 into 1 European swaption vol in a year. Consider again our $100mm 1 into 2 Bermudan payer swaption struck at 5% that we are long. Suppose that in one year the swap curve is as in Table 5.9. Suppose further that one year from now 1 into 1 European swaption vol is 25%, as compared to the level of 20%, which was the level of 1 into 1 swaption vol assumed to be prevailing in one year in Example 2. Is it optimal to exercise our Bermudan payer swaption at this time?

The 2-year swap rate prevailing at this time is still 5.30%. As before, if we exercise at this point, we will be paying fixed in a 2-year swap at a rate of 5% when the 2-year swap rate is 5.30%. As before, the value from exercising is $555,461.

If, however, we do not exercise our Bermudan swaption, we will again be long the residual option in the Bermudan, which is a 1 into 1 European payer swaption struck at 5%. The valuation of this option is very similar to the calculation in Example 2, but now we assume that implied 1 into 1 vol is

Swaption model		1	into	1		
Time	Swap rate	zc	Period	$DF(0,t)$		
1	5.300%	5.300%	0x1		d_1	0.3913
2	5.300%	5.300%	1x2	0.9019	d_2	0.1913
3	5.300%	5.300%	2x3			
4	5.300%	5.300%	3x4		$FN(d_1)$ - $XN(d_2)$	0.005775
					Payer swaption	520,785
					(in bps)	52.1
F	5.300%		A =	0.9019		
σ	20.0%					
T	1		Notional	100,000,000		
X	5.000%					

Swaption model		1	into	1		
Time	Swap rate	zc	Period	$DF(0,t)$		
1	5.300%	5.300%	0x1		d_1	0.3581
2	5.300%	5.300%	1x2	0.9019	d_2	0.1081
3	5.300%	5.300%	2x3			
4	5.300%	5.300%	3x4		$FN(d_1)$ - $XN(d_2)$	0.006761
					Payer swaption	609,735
					(in bps)	61.0
F	5.300%		A =	0.9019		
σ	25.0%					
T	1		Notional	100,000,000		
X	5.000%					

Figure 5.18: Valuation of 1 into 1 European Payer Struck at 5% when the Swap Curve is as in Table 5.9 and 1 into 1 Vol is 20% (*top half*) and 1 into 1 Vol is 25% (*bottom half*)

Example	2-Year Swap Rate in a Year	σ_{LN}	Exercise Early?
1	5.01%	20%	No
2	5.30%	20%	Yes
3	5.30%	25%	No

Table 5.10: Summary of Early Exercise Decision for the 1 into 2 Bermudan Payer Struck at 5% Considered in Examples 1-3

25%, instead of 20%. As shown in the bottom half of Figure 5.18, the value of this swaption is \$609,735. In this case, we are better off not exercising at the first exercise date since the value of the residual option exceeds the value obtained from exercising. □

Table 5.10 summarizes what we have found in these three examples. Example 1 demonstrated that if the then-prevailing 2-year swap rate is above the strike in a year, this does not necessarily mean that we should exercise the Bermudan at the first exercise date. If the then-prevailing 2-year swap rate is only slightly above the strike, we may be better off forgoing exercise at the first exercise date in order to maintain a long position in the residual option. Example 2 shows that there are cases in which the then-prevailing 2-year swap rate in a year is high enough above the strike so to make exercise at the first exercise date optimal. Example 3 shows that the decision made at the first exercise date is not purely a function of the level of rates prevailing in a year; the higher the level of 1 into 1 swaption vol prevailing in a year, the greater the value that is attributed to holding on to the residual 1 into 1 payer swaption we maintain by forgoing exercise at the first exercise date.

Example 4

Consider once again our stylized example and suppose we purchase \$100mm of a 1 into 3 year 5% Bermudan payer swaption. This option gives us the right to exercise in a year and pay fixed in a 3-year swap or to exercise in two years and pay fixed in a 2-year swap or to exercise in three years and pay fixed in a 1-year swap.

Suppose now one year has gone by. If we exercise today, we will pay fixed in a 3-year swap at a rate of 5%. If we don't exercise today, our long position will be a 1 into 2 5% Bermudan payer swaption. As in the previous examples we have considered in this section, our exercise decision will depend upon which one of these two is greater in value. It is straightforward to compute the value of the swaption that results from exercise today. This is simply the present value of an annuity equal to the difference between the prevailing 3-year swap rate and the strike of 5% for three years. However, if we do not exercise today, we remain long a 1 into 2 Bermudan payer struck at 5%. Thus, our exercise decision is complicated by the fact that it depends on the valuation of a (residual) Bermudan swaption. The valuation of the 1 into 2 5% Bermudan is beyond the scope of this discussion. □

The first three examples is this section considered the relatively simple case of a Bermudan swaption in which the remaining option at the first exercise date is a simple European. These cases were relatively easy to analyze. Example 4 starts to reveal the complexity inherent in Bermudan swaptions; even the exercise of a Bermudan swaption depends on the valuation of the residual swaption position, which in general may be a Bermudan option itself.

5.6.2 Valuation of Bermudan Swaptions

The multiple exercise dates of a Bermudan swaption make this an exotic option. Although exotic options are often regarded as illiquid, the ubiquity of this type of option in the market has made it a relatively liquid product.[43] In fact, it has become very common for many market participants to traffic in Bermudan swaptions, sometimes buying these exotic structures instead of Europeans if they are trading at relatively cheap levels.[44]

The analysis and pricing of Bermudan swaptions is very complicated. The use of Black's formula is restricted to options that have only a single exercise date. When a Bermudan swaption is first purchased, we are not certain on which exercise date (if any) the option will be exercised; there is some probability associated with exercise at each of the exercise dates (and of course some probability that the option will be exercised at none of the exercise dates). To further complicate things, as we will see the analysis of Bermudans is intimately linked to the consideration of skew.

The rigorous valuation of Bermudan swaptions is beyond the scope of this discussion.[45] That being said, we will present a heuristic paradigm for pricing Bermudans as a portfolio of European swaptions, based on an elegant methodology provided by Zhou (2003). Consider now a Bermudan payer swaption that has precisely two exercise dates. This Bermudan option gives the holder the right to pay fixed at a strike rate X. This option may be exer-

[43] In Chapter 6, we will discuss one of the common uses of Bermudan swaptions.

[44] Trading Bermudan swaptions versus their close counterparts, European swaptions, has become a popular relative value trade in which traders express their view on the relative richness or cheapness of one type of option versus the other.

[45] For various methodologies used to price Bermudan swaptions, see, for example, Brigo and Mercurio (2006), Andersen (2000), Pedersen (1999), Andersen and Andreasen (2001), and Longstaff and Schwartz (2001).